ARJUNA JEE 2027

Lecture -09 | Mathematics

BINOMIAL THEOREM

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Topics *to be covered*

A **Binomial Theorem for Rational Index**

B **Use of Multinomial Theorem**

C **Question Practice**

D



H.W. Discussion

Question [JEE Mains-2023]

The remainder, when $19^{200} + 23^{200}$ is divided by 49, is _____.

$49 = 7^2$

$$(21-2)^{200} + (21+2)^{200}$$

$$2\left\{{}^{200}C_0\, 21^{200} + {}^{200}C_2\, 21^{198}\, 2^2 + {}^{200}C_4\, 21^{196}\, 2^4 + \ldots\ldots + {}^{200}C_{198}\, 21^2\, 2^{198} + {}^{200}C_{200}\, 2^{200}\right\}$$

$$2 \times 49(I) + 2^{200} \times 2^1$$

$$\frac{2^{201}}{49} = \frac{(2^3)^{67}}{49} = \frac{(7+1)^{67}}{49}$$

7^2

$$fnf = 29$$

$$(7+1)^{67} = {}^{67}C_0\,7^{67} + {}^{67}C_1\,7^{66} + \ldots\ldots\ldots + {}^{67}C_{65}\,7^2 + {}^{67}C_{66}\,7^1 + {}^{67}C_{67}$$

$49(I) + 67\times 7 + 1$

$$\underbrace{67\times 7}_{49} + \underbrace{1}_{49}$$

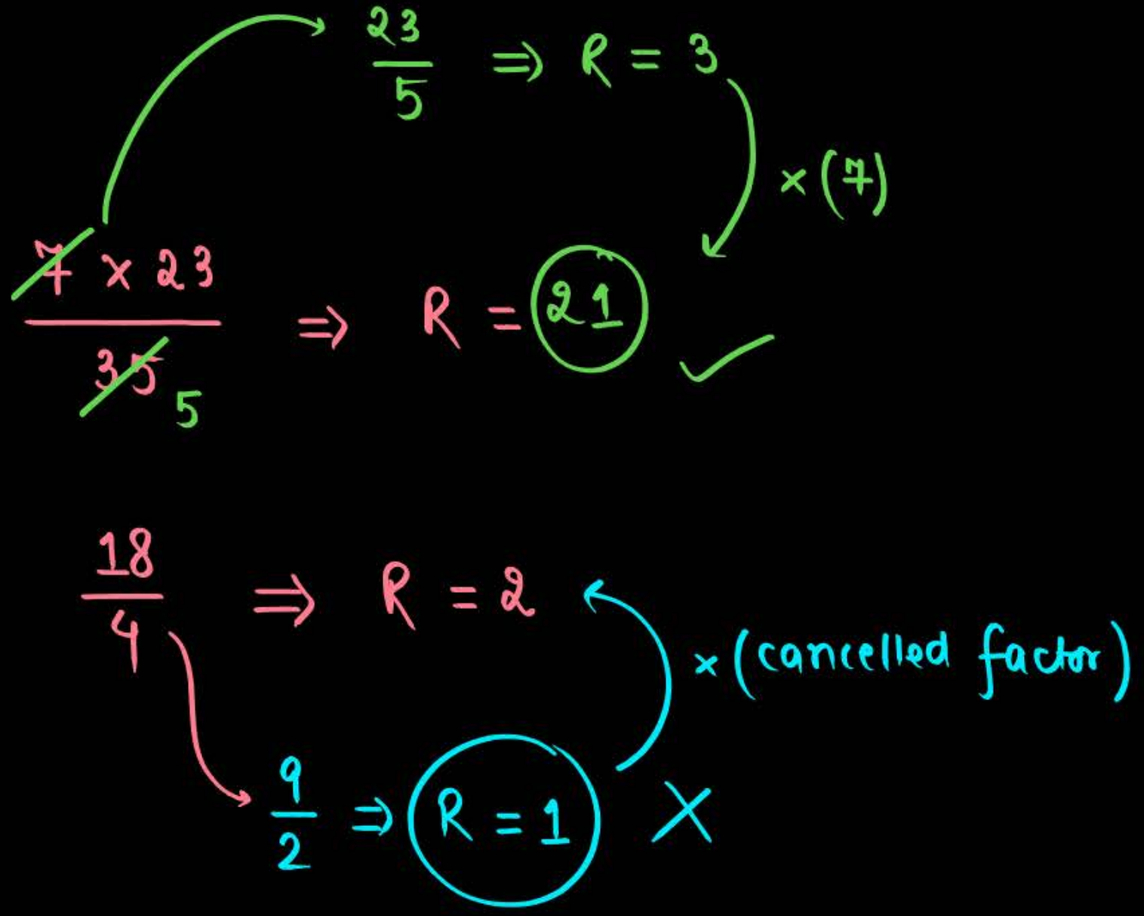
$R = 28 + 1$

$$(x+y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} y^1 + {}^nC_2 x^{n-2} y^2 + \dots\dots\dots$$

$$(x-y)^n = {}^nC_0 x^n - {}^nC_1 x^{n-1} y^1 + {}^nC_2 x^{n-2} y^2 - \dots\dots\dots$$

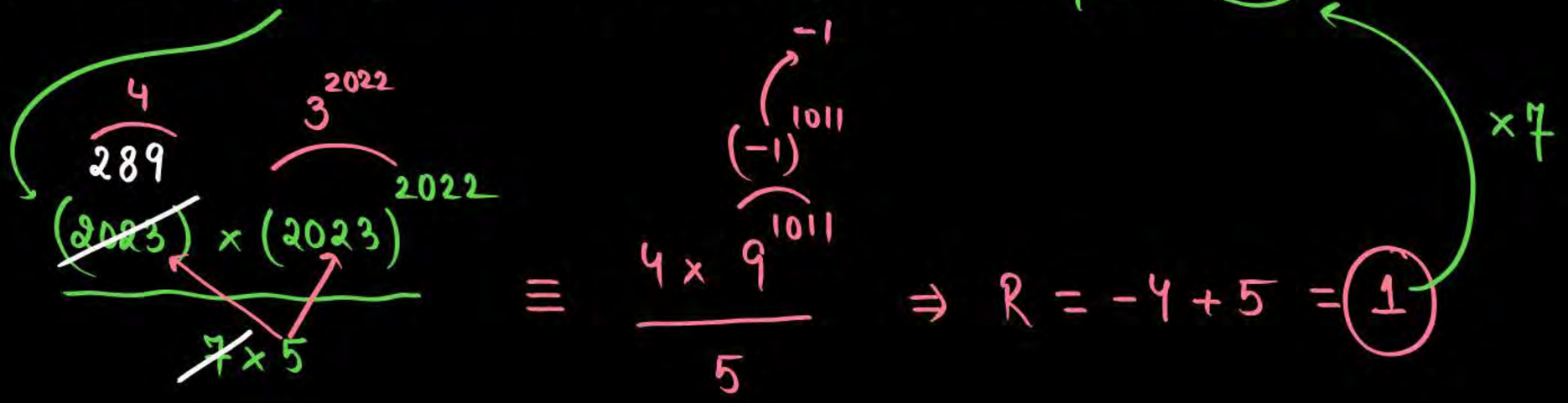
add

$$(x+y)^n + (x-y)^n = 2\left\{ {}^nC_0 x^n + {}^nC_2 x^{n-2} y^2 + \dots\dots \right\}$$


$\frac{23}{5} \Rightarrow R = 3$
$\times (7)$
$\frac{7 \times 23}{35_5} \Rightarrow R = 21$
$\frac{18}{4} \Rightarrow R = 2$
$\times$ (cancelled factor)
$\frac{9}{2} \Rightarrow R = 1$
X

Question [JEE Mains-2023]

The remainder when $(2023)^{2023}$ is divided by 35 is _____.



Question [JEE Mains-2023]

$x^n - y^n$ is always div. by $(x-y)$

Among the statements :

(S1) : $2023^{2022} - 1999^{2022}$ is divisible by 8 → div. by $(2023-1999) = 24$

(S2) : $13(13)^n - 11n - 13$ is divisible by 144 for infinite many $n \in N$.

A. Both (S1) and (S2) are incorrect

B. Only (S1) is correct

C. Only (S2) is correct

D. Both (S1) and (S2) are correct

$13(12+1)^n - 11n - 13$

$13\left({}^nC_0 12^n + {}^nC_1 12^{n-1} + \ldots\ldots + {}^nC_{n-2} 12^2 + {}^nC_{n-1} 12^1 + {}^nC_n\right) - 11n - 13$

$13 \times 144(I) + n \times 12 \times 13 + 13 - 11n - 13$

$144n + n$ → $144\times1, 144\times2, 144\times3, \ldots$

Question [JEE Mains-2024]

Product of last two digits of 1919^{1919} is _____.

63 ✓

$(1920-1)^{1919}$

$${}^{1919}C_0(1920)^{1919} - {}^{1919}C_1(1920)^{1918} + \ldots\ldots - {}^{1919}C_{1917}(1920)^2 + {}^{1919}C_{1918}(1920)^1 - {}^{1919}C_{1919}$$

$100(I) + 1919 \times 1920 - 1$

Done!!

Question

If $(\sqrt{17}+4)^7 = I + f$, then show $I = \frac{1}{f} - f$.

$$(\sqrt{17}+4)^7 = {}^nC_0(\sqrt{17})^n + {}^nC_1(\sqrt{17})^{n-1}\cdot 4^1 + {}^nC_2(\sqrt{17})^{n-2}\cdot 4^2 + \text{----------}$$

$$(\sqrt{17}+4)^7 = {}^7C_0(\sqrt{17})^7 + {}^7C_1(\sqrt{17})^6 \cdot 4^1 + {}^7C_2(\sqrt{17})^5 \, 4^2 + \dots\dots\dots$$

$$(\sqrt{17}-4)^7 = {}^7C_0(\sqrt{17})^7 - {}^7C_1(\sqrt{17})^6 \cdot 4^1 + {}^7C_2(\sqrt{17})^5 \cdot 4^2 - \dots\dots\dots$$

$\ominus$

$$(\sqrt{17}+4)^7 - (\sqrt{17}-4)^7 = 2\left\{ {}^7C_1(\sqrt{17})^6 \cdot 4^1 + {}^7C_3(\sqrt{17})^4 \cdot 4^3 + \dots\dots\dots \right\}$$

$$I + f - G = 2(I)$$

integer.

$\hookrightarrow f - G = 0$

$f = G$.

$$(\sqrt{17}+4)^7 \cdot (\sqrt{17}-4)^7 = (I+f)G$$

$\hookrightarrow 1 = (I+f)f$

$\frac{1}{f} - f = I$ ✓

B.T. FOR RATIONAL NUMBER & -VE INTEGERS

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$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots\ldots\infty$$

This expression is valid only if $|x| < 1$



ex: $(1+x)^{\frac{4}{3}}$, $(1-x)^{-3}$, ------

RAPID FIRE ROUND

(i) Fourth term in the expansion $(1-2x)^{\frac{3}{2}}$

(ii) Value of x for which $\frac{1}{\sqrt{5+4x}}$ can be expanded as ∞ series. in ↑ powers of x.

(iii) Using binomial calculate $\sqrt{10}$ upto two decimal places.

(iv) Using binomial calculate $(121)^{\frac{1}{3}}$ upto two decimal places.

(i) $T_4 = \frac{n(n-1)(n-2)}{3!_0}x^3 = \frac{\left(\frac{3}{2}\right)\left(\frac{3}{2}-1\right)\left(\frac{3}{2}-2\right)}{3!_0}(-2x)^3$ ✓

(ii) $\frac{1}{\sqrt{5+4x}} = (5+4x)^{-\frac{1}{2}} = 5^{-\frac{1}{2}}\left(1+\frac{4x}{5}\right)^{-\frac{1}{2}} \Rightarrow \left|\frac{4x}{5}\right| < 1 \Rightarrow \frac{4}{5}|x| < 1 \Rightarrow |x| < \frac{5}{4}$

$-\frac{5}{4} < x < \frac{5}{4}$ ✓

RAPID FIRE ROUND

(i) Fourth term in the expansion $(1-2x)^{\frac{3}{2}}$

(ii) Value of x for which $\frac{1}{\sqrt{5+4x}}$ can be expanded as ∞ series.

(iii) Using binomial calculate $\sqrt{10}$ upto two decimal places.

(iv) Using binomial calculate $(121)^{\frac{1}{3}}$ upto two decimal places.

(iv) $(125-4)^{\frac{1}{3}} = 5\left(1-\frac{4}{125}\right)^{\frac{1}{3}}$

(iii) $(10)^{\frac{1}{2}} = (9+1)^{\frac{1}{2}} = 3\left(1+\frac{1}{9}\right)^{\frac{1}{2}}$

$= 3\left(1+\frac{1}{2}\times\frac{1}{9}+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\frac{1}{81}+\ldots\ldots\infty\right)$

$= 3+\frac{1}{6} \simeq 3.16$

$\sqrt{7} = (9-2)^{\frac{1}{2}} = 3\left(1-\frac{2}{9}\right)^{\frac{1}{2}}$

OR

$= (4+3)^{\frac{1}{2}} = 2\left(1+\frac{3}{4}\right)^{\frac{1}{2}}$

IMPORTANT NOTE

- $(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \cdots \infty$
- $(1-x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \cdots \infty$
- $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \cdots \infty$
- $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \cdots \infty$
- $(1+x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \cdots \infty$
- $(1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \cdots \infty$

S & S

∞ – G.P.

$$\frac{1}{1-(-x)} = \frac{1}{1+x} = (1+x)^{-1}$$

B.T.

$$(1+x)^{-1} = 1 + (-1)x + \frac{(-1)(-1-1)}{2!}x^2 + \frac{(-1)(-1-1)(-1-2)}{3!}x^3 + \cdots \infty$$

$$= 1 - x + x^2 - x^3 + x^4 - \cdots \infty$$

When power is negative integer:

$n \in N$

$$(1-x)^{-n} = 1 + (-n)(-x) + \frac{(-n)(-n-1)}{2!}(-x)^2 + \frac{(-n)(-n-1)(-n-2)}{3!}(-x)^3 + \ldots\ldots\infty$$

$$= 1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \ldots\ldots\infty$$

nC_1 $\quad$ $n+1C_2$ $\quad$ $n+2C_3$

Coefficient of x^r in $(1-x)^{-n} = {}^{n+r-1}C_r$

Coefficient of x^r in $(1-x)^{-n}$ = $^{n+r-1}C_r$

RAR

Coeff. of x^7 in $(1-x)^{-3}$ is $^{3+7-1}C_7 = {}^9C_7$.

Coeff. of x^r in $(1+x)^{-n}$ is $^{n+r-1}C_r\,(-1)^r$

Coeff. of x^7 in $(1+x)^{-3}$ is $^{3+7-1}C_7\,(-1)^7 = -{}^9C_7$.

Coefficient of x^r in $(1-x)^{-n} = {}^{n+r-1}C_r$

RAR

Coeff. of x^7 in $(1-2x)^{-3}$ is ${}^{3+7-1}C_7 2^7 = {}^{9}C_7 \cdot 2^7$.

$$1 + 3(2x) + \frac{(-3)(-3-1)}{2!} 2^2 x^2 + \frac{(-3)(-3-1)(-3-2)}{3!}\left(-2^3 x^3\right) + \dots$$

Question [JEE Mains-2019]

The coefficient of t^4 in the expansion of $\left(\frac{1-t^6}{1-t}\right)^3$ is

A 15

B 14

C 12

D 10

$$\frac{(1-t^6)^3}{(1-t)^3} = (1-t^6)^3 \times (1-t)^{-3}$$

$$\left({}^3C_0\right)t^0 \times \left({}^6C_4\right)t^4$$

$${}^{3+4-1}C_4$$

$${}^6C_4 \rightarrow 15$$

$${}^3C_r(-t^6)^r$$

$r=0 \Rightarrow {}^3C_0$

$15t^4$

Question

The coefficient of x^n in $\frac{(1+x)^2}{(1-x)^3}$ is

A $3n^2 + 2n + 1$

B $2n^2 + 2n + 1$

C $n^2 + n + 1$

D $2n^2 - 2n + 1$

$$(1+x^2+2x)(1-x)^{-3}$$

$$(1)x^0 \times {}^{3+n-1}C_n \; x^n$$

$$(2)x^1 \times {}^{3+n-2}C_{n-1} \; x^{n-1}$$

$$(1)x^2 \times {}^{3+n-3}C_{n-2} \; x^{n-2}$$

APPLICATION OF MULTINOMIAL THEOREM

P&C (Number of Integral Solutions):

Question

In how many ways sum of 12 can be obtained by throw of 4 distinct dices.



$a + b + c + d = 12$, $a, b, c, d \in \{1, 2, 3, 4, 5, 6\}$

$\times$ 9 1 1 1

$\times$ 7 3 1 1

$\times$ 1 2 8 1

$\vdots$

$\eta.o.s. = {}^{12-1}C_{4-1} = {}^{11}C_3$. $\times$



$a + b + c + d = 12$, $a, b, c, d \in \{1, 2, 3, 4, 5, 6\}$

Coeff. of x^{12} in $(x^1+x^2+x^3+\ldots+x^6)(x^1+x^2+x^3+\ldots+x^6)(x^1+x^2+x^3+\ldots+x^6)(x^1+x^2+x^3+\ldots+x^6)$

Coeff. of x^{12} in $(x^1+x^2+x^3+\ldots+x^6)^4$.

Coeff. of x^{12} in $\left(x\left(\frac{1-x^6}{1-x}\right)\right)^4$ $\Rightarrow$ Coeff. of x^{12} in $x^4\frac{(1-x^6)^4}{(1-x)^4}$

Coeff. of x^8 in $\rightarrow \frac{(1-x^6)^4}{(1-x)^4}$

Coeff. of x^8 in $\rightarrow \dfrac{(1-x^6)^4}{(1-x)^4}$

$(1-x^6)^4$

$\hookrightarrow {}^4C_r (-1)^r \cdot x^{6r}$

Coeff. of x^8 in $\rightarrow (1-x^6)^4 (1-x)^{-4}$

$({}^4C_0)x^0 \cdot ({}^{4+8-1}C_8)x^8 = {}^{11}C_8\, x^8$

$(-{}^4C_1)x^6 \cdot ({}^{4+2-1}C_2)x^2 = -{}^4C_1\, {}^5C_2\, x^8$

$= ({}^{11}C_8 - 4\times 10)x^8$

$= 165 - 40 = 125$



$a + b + c + d = 10$, $0 \le a \le 3$, $2 \le b \le 5$, $2 \le c \le 5$, $d \ge 0$

Coeff. of x^{10} in $\left(x^0+x^1+x^2+x^3\right)\left(x^2+x^3+x^4+x^5\right)\left(x^2+x^3+x^4+x^5\right)\left(x^0+x^1+x^2+x^3+\ldots\ldots\right)$

Question [JEE Mains-2022]

The number of ways to distribute 30 identical candies among four children C_1, C_2, C_3 and C_4, so that C_2, receives atleast 4 and atmost 7 candies, C_3 receives atleast 2 and atmost 6 candies, is equal to

$C_1 \geq 0$ $\quad$ $C_4 \geq 0$

$$C_1 + C_2 + C_3 + C_4 = 30.$$

$4 \leq C_2 \leq 7$ $\quad$ $2 \leq C_3 \leq 6$



A 205

B 615

C 510

D 430



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

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